Assembling the IN-16 sockets

I don't want to give advice to anyone about soldering nixie tubes, in this document I just show how it works for me.

Nixies are expensive these days and are the most valuable parts of this project, so avoiding mistakes is important.

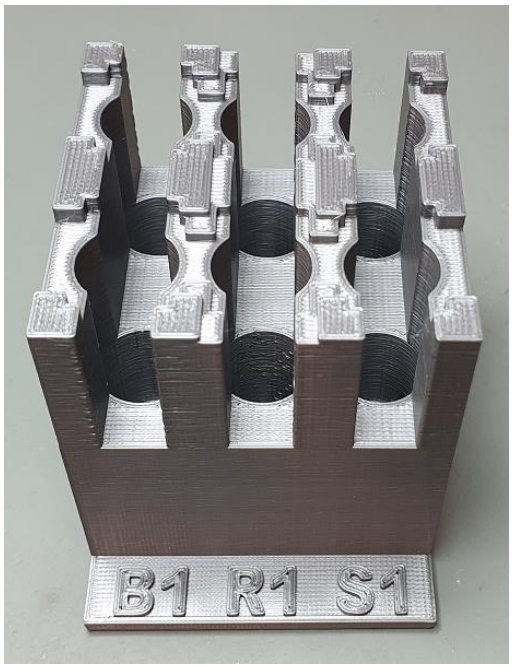
WARNING: The dimensions of tubes of the same type can vary considerably. Don't use nixies wider than 13mm for this project if you cannot compensate the difference with the adjacent tubes! Also check, that the exhaust tips are not too long to fit in the case.

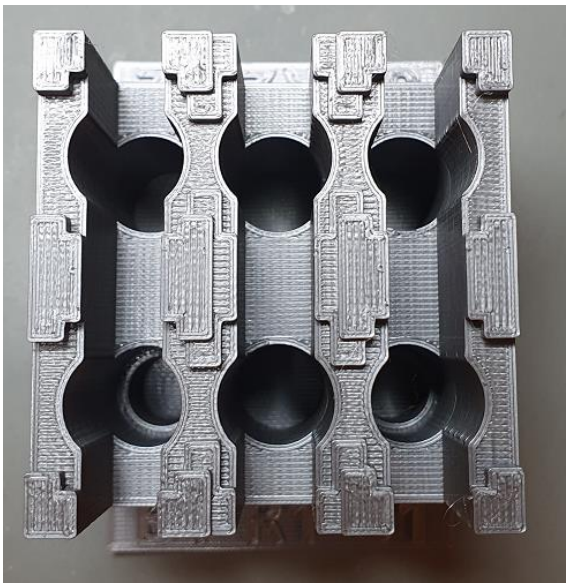
I didn't have any problems with any of my (non-matched) tubes.

It is not that easy to align 14 tubes, so I ended up with a rather "complicated" method with a 3D printed auxiliary tool. Of course, if you prefer, you can do it by eye.

The following steps worked for me:

1. Print the IN-16 tool



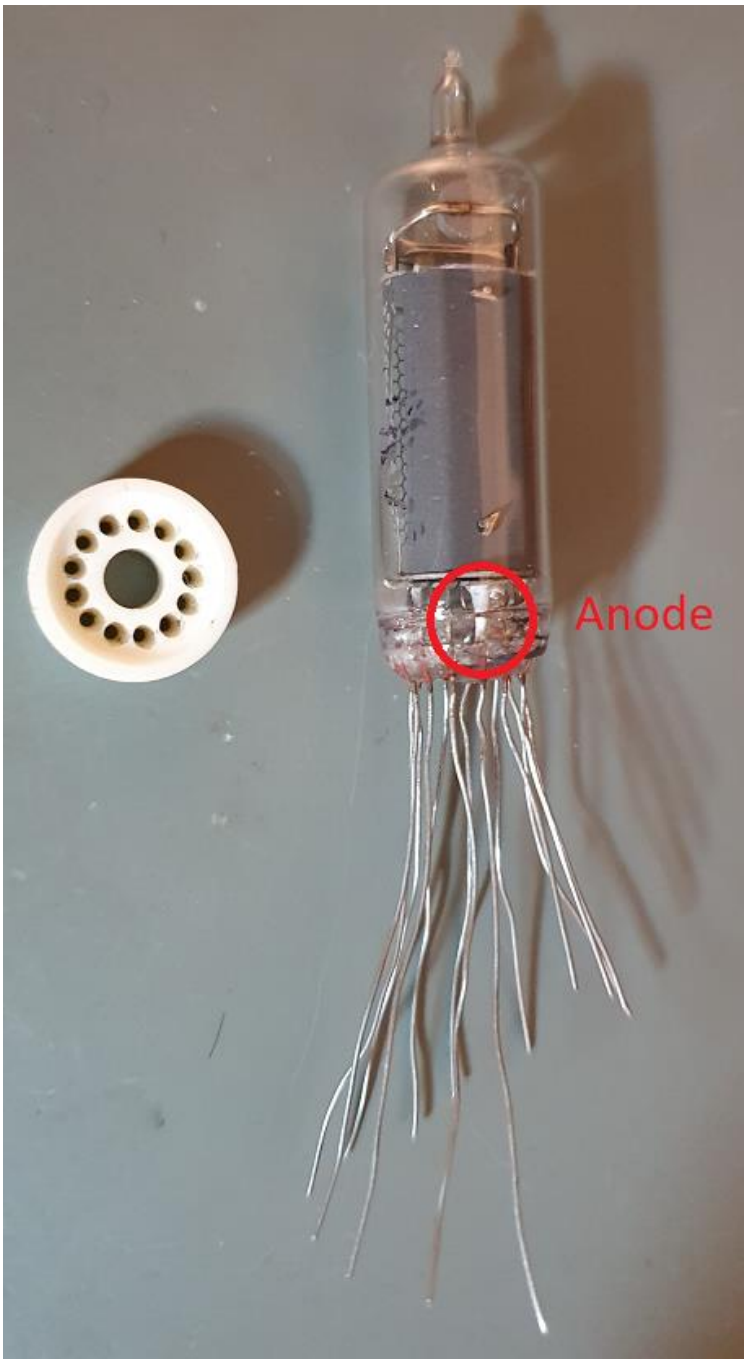


2. Start assembling the calculator and stop **after** the step “Install the driver and display boards” (see Docs/Assembling the IN-16 calculator.pdf) This is important because we can check if the reference tube fits into the case.

3. Take a spare IN-16 display board and 2 spare socket headers and solder the header H1_1 and H1_2 to the board. If you want to test the nixies you can also populate R1 and R2.

Now we are ready to build the reference tube socket. Of all the IN-16s that you plan to use for this project, take the one with the longest exhaust tip.

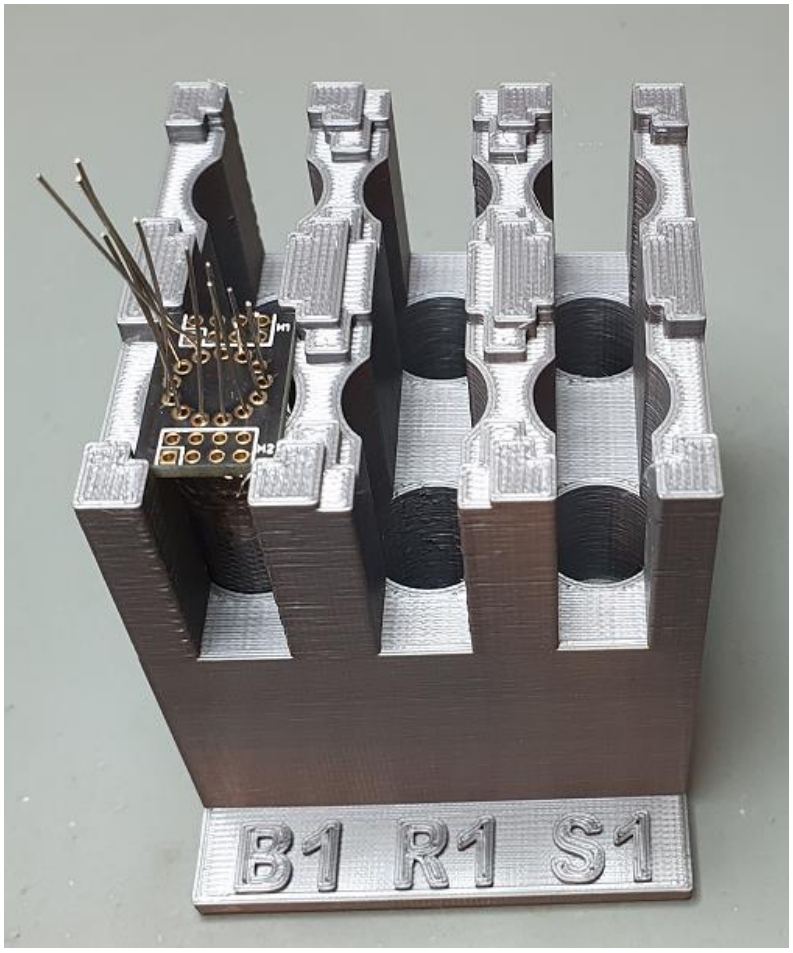
4. Carefully remove the plastic spacer and find the anode lead (**double-check with IN-16 datasheet!**) Leave the anode lead as it is, take the next (adjacent) lead and cut about 1mm from the free end of the lead. Then take the next lead and cut 2mm, take the next lead and cut 3mm and so on…



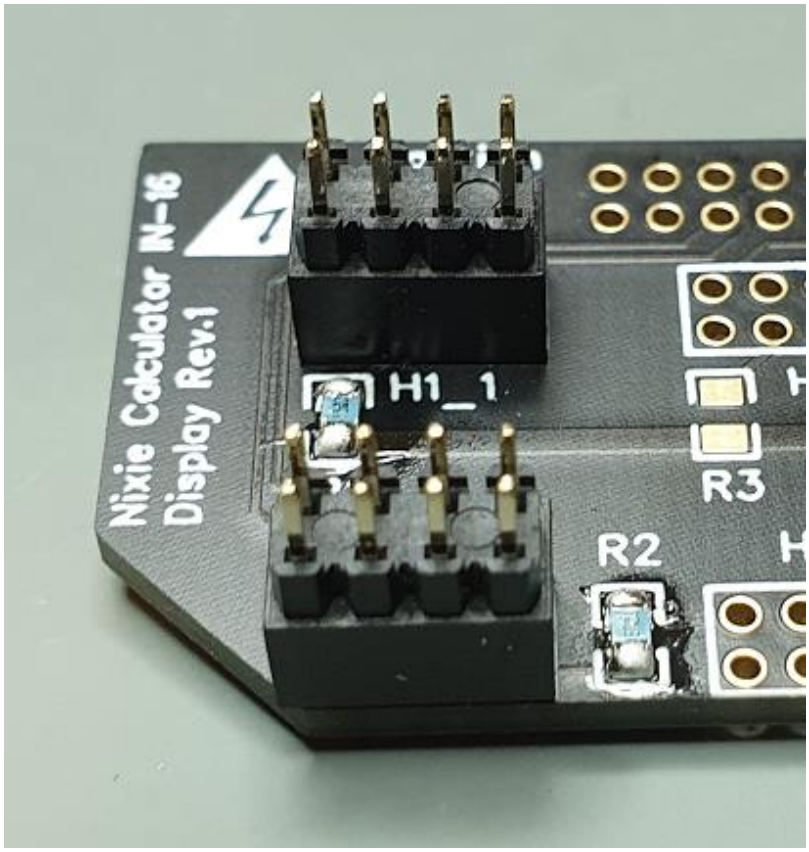
5. Introduce the leads in a socket board starting with the anode lead. On the board the anode pad is marked with an A

6. Bring the socket board into position. Try first slot **S1**. If the nixie does not fit, try slot **B1**. **Do not use R1, R2, S2 or B2 for the reference tube, they don't have the inner ring for the reference height.**

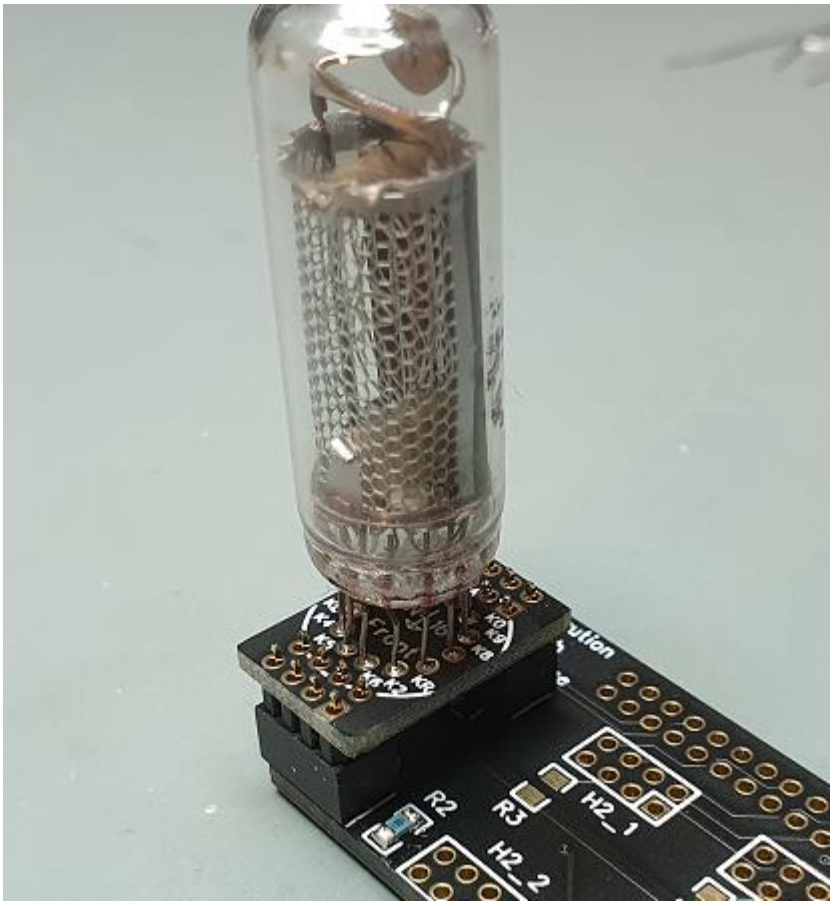
7. **Push the tube all the way down**. Make sure everything is correct, then solder the nixie and finally cut the leads.



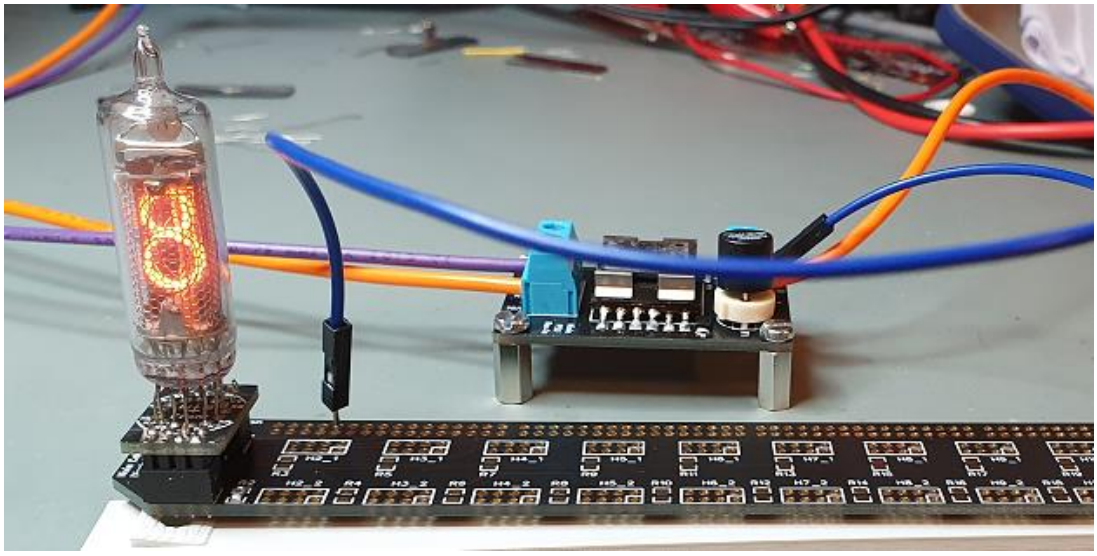
8. Put headers (longer pins facing down) into the sockets on the display board



9. Place the nixie socket on the headers and solder the pins

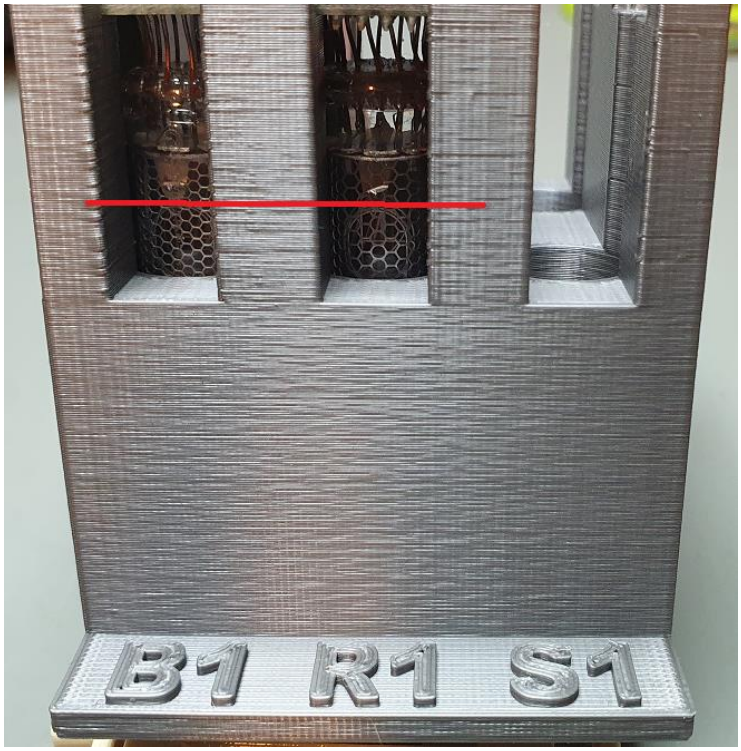


10. You can now test the nixie

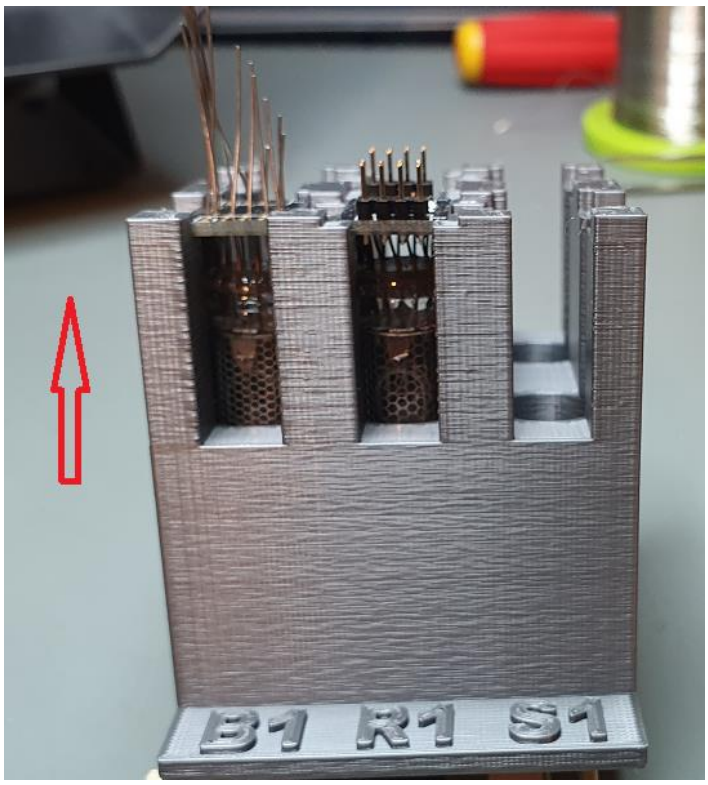
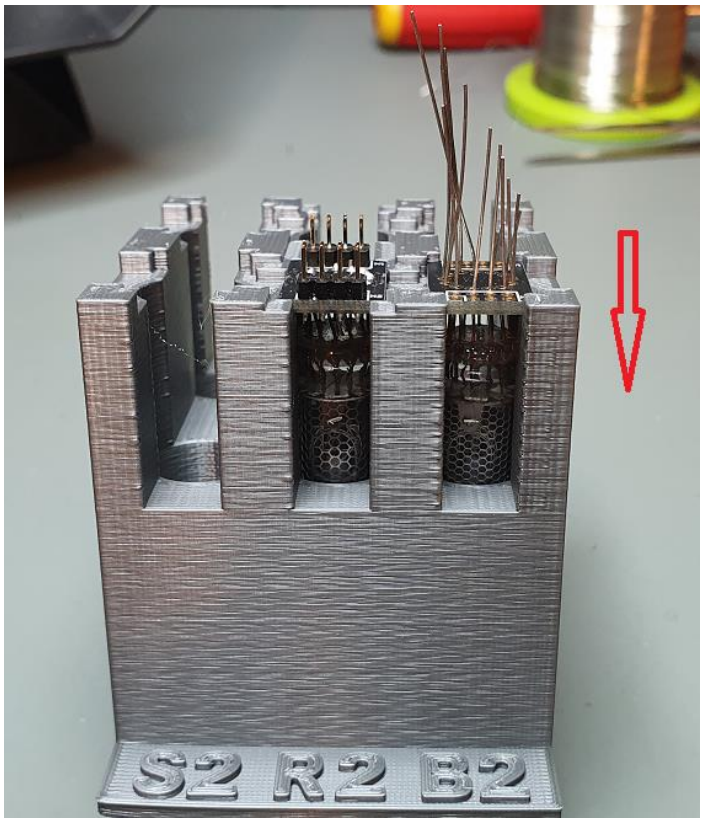


11. **Check if the nixie with the socket fits into the case.** If everything is fine, this is now your reference tube.

12. Put the reference tube into slot R1 and start with the next socket. Before you solder, you can now visually check if the digits are on the same level.



If the digit is too low, you can just lift the tube to adjust the level.



If the digit is too high, you have to switch to the “2” slots and push the tube down to adjust the level.